

PROFESSIONAL SUMMARYAspiring AI and Software Engineer

CSE undergraduate at IIIT Delhi (CGPA 8.9) skilled in Python, C++, React.js, and FastAPI, with exposure to NumPy, Pandas, and OpenCV for AI/ML workflows. Interested in Artificial Intelligence and Machine Learning, Software Development and Research. Have experience in building real world technical project with deployment.

WORK EXPERIENCE & LEADERSHIP

- BYLD CLUB- BYLD is the esteemed software development club at IIITD

- Team Lead- Recently promoted to the lead position

- Team Member-

Organized and managed technical hackathons and innovation challenges.

Conducted workshops on Git & GitHub, helping students develop version control, collaboration, and open-source contribution skills.

Organized and facilitated an n8n Automation workshop for students

May 2026 - Present

August 2025- present
- CiPD(Centre for Intelligent Product Development)- A research and innovation hub

focused on intelligent electronic product development.

Organized a hands-on robotics workshop focused on designing, assembling, and operating mini battle robots for RoboWar competitions.

Coordinated a Product-A-Thon, encouraging participants to ideate, design, and develop innovative technology-driven product solutions.

September 2025- present
- Undergraduate Research Club- Fosters AI/ML research through collaborative learning,

mentorship, and research groups.

Recently recruited in the club.

April 2026- present

EDUCATIONCBSE X: 96.8% (2023) | CBSE XII: 93.4% (2025)

IIITD (Indraprastha Institute Of Information and Technology)2025-2029

B.Tech in Computer Science and Engineering (CSE) | CGPA-8.9

- RELEVANT COURSEWORK (CGPA)
- Intro to Programming (python) - 9

Data Structure & Algorithms (C++) - 9

probability & Statistics - 8

Digital Circuits - 9

Basic Electronics - 9

Linear Algebra - 10

Human Computer Interaction(HCI) - 10

Computer Organisation - 9

PROJECTS:

- Delhi Metro Route and Schedule Simulator- <https://github.com/rudrakshia30/Delhi-Metro-Route-and-Schedule-Simulator>

Developed a Python-based metro simulation system using real DMRC data for route planning and train scheduling.

Implemented optimal route calculation, next train arrival prediction, travel time estimation, and dynamic fare computation.

Integrated Blue, Magenta, and Red line support with interchange handling and peak/off-peak scheduling using file handling without external libraries.

RISC-V Assembler project.- <https://github.com/rudrakshia30/RISC-V-assembler>

Built an RV32I assembler in Python to convert assembly instructions into 32-bit binary machine code.

Implemented support for R, I, S, B, U, and J-type instruction formats with label resolution and PC-relative offset calculations.

Added comprehensive error handling for invalid syntax, registers, immediates, empty lines, and missing halt conditions with line-specific reporting.

RISC-V Simulator project.- <https://github.com/rudrakshia30/RISC-V-simulator>

Developed a Python-based RV32I simulator capable of executing 32-bit binary machine code instructions.

Simulated registers, memory operations, branching logic, and program counter updates for complete instruction execution.

Implemented instruction tracing, memory dumps, alignment checks, infinite loop detection, and runtime error handling for debugging and analysis.

SKILLS:

Languages: Python, C++, C, JavaScript

Frameworks & Technologies: React.js, FastAPI, HTML5, CSS3

AI/ML & Computer Vision: NumPy, Pandas, Matplotlib, Seaborn, OpenCV, LLM Integration

Developer Tools: Git, GitHub, VS Code, Microsoft Excel, Figma, Canva

Core Concepts: Data Structures & Algorithms, OOP, Software Development, Web Development, Problem Solving

Professional Skills: Leadership, Teamwork, Communication, Time Management, Collaboration